Clinical Risk Assessment Report: Patient 76

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 34.0% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 1.54x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM_LGZE.CT), which elevated the patient's absolute risk by +6.9%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.32 [Bottom 19%] (+6.9%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 0.5 [Bottom 10%] (+3.3%)**

Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 0.06 [Avg (51th %ile)] (+3.0%)**

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.19 [Avg (27th %ile)] (+0.9%)**

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT) **Value: -1.07 [Bottom 14%] (+0.7%)**

Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

Tumor Sphericity (PET) (SHAPE_Sphericity(onlyFor3DROI)).PET) **Value: 0.32 [Avg (57th %ile)] (+0.7%)**

Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

Minimum Tumor Density (CONVENTIONAL_HUmin) **Value: 0.79 [Top 21%] (+0.7%)**

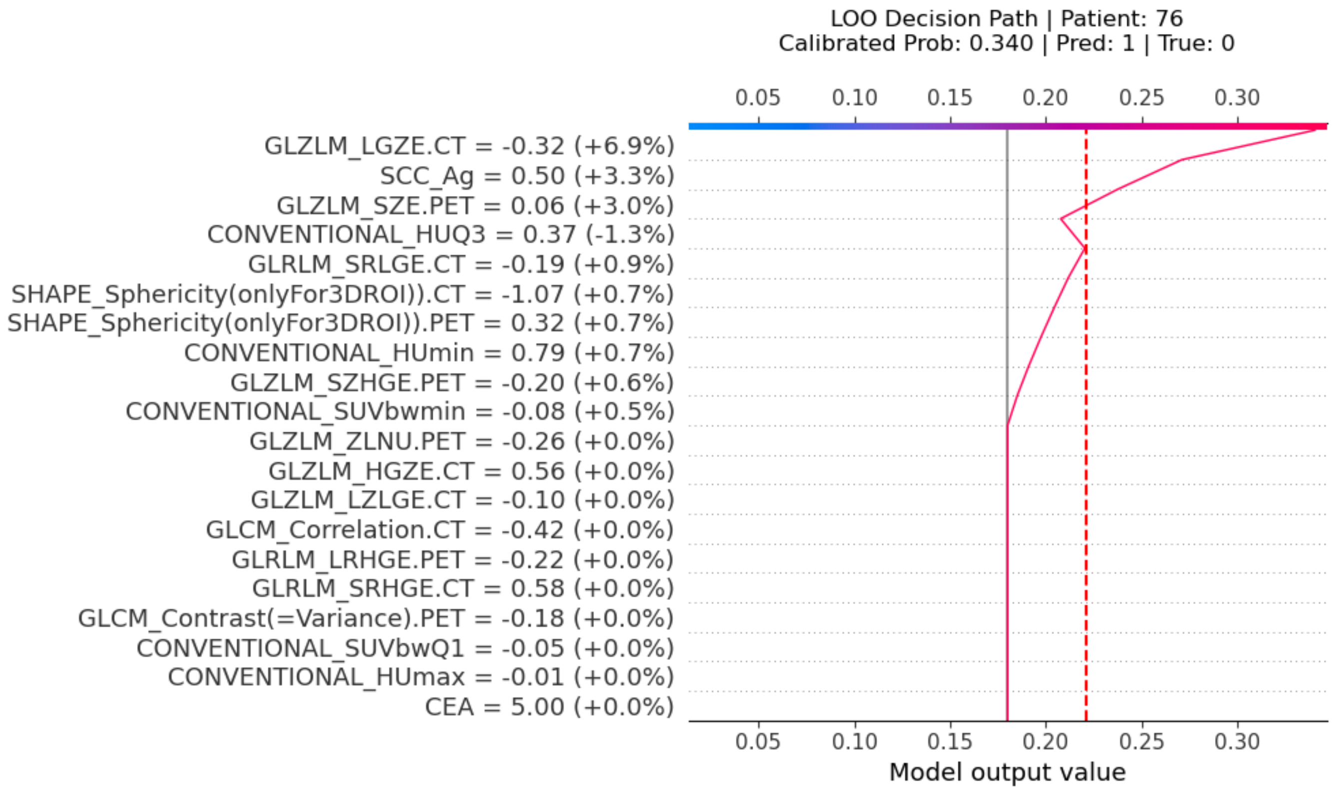
Reflects the least dense areas within the tumor on CT, often corresponding to cavitation, necrosis, or cystic degeneration.

3. Protective / Mitigating Factors (Reducing Risk)

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.37 [Avg (54th %ile)] (-1.3%)**

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Machine Learning Feature Attribution (SHAP Decision Path)



Sensitivity Analysis: Feature Tipping Points

The plots below demonstrate how variations in specific clinical or radiomic features affect the model's predicted risk. The red marker indicates the patient's current clinical state. Crossing the red dashed line changes the diagnosis.